

FLORENCE, SC, USA

WMO: 723106

Lat: 34.188N

Lon: 79.731W

Elev: 45

StdP: 100.79

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13744

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.7	-2.8	-14.7	1.1	0.4	-12.1	1.3	1.9	10.6	12.4	9.1	11.9	2.2	340	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	35.5	24.7	34.2	24.4	33.0	24.2	26.3	32.4	25.7	31.5	25.3	30.7	3.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.9	28.4	24.2	19.2	27.9	23.8	18.8	27.6	81.6	32.3	79.3	31.7	77.3	30.7	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.8	6.9	DB	-9.2	37.3	2.5	1.7	-11.0	38.5	-12.5	39.5	-13.9	40.4	-15.7	41.6	(4)
(5)			WB	-10.1	27.4	2.3	0.9	-11.8	28.0	-13.1	28.6	-14.4	29.0	-16.1	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	7.7	9.6	13.2	17.8	22.2	26.0	27.5	26.8	24.0	18.5	12.7	9.4	(6)	
	DBStd	8.13	5.81	5.30	5.03	4.16	3.50	2.61	1.99	2.08	2.93	4.15	4.68	5.31	(7)	
	HDD10.0	314	115	68	27	2	0	0	0	0	0	1	24	77	(8)	
	HDD18.3	1329	330	249	170	59	9	1	0	0	2	50	178	281	(9)	
	CDD10.0	3231	44	56	127	237	377	479	542	522	419	264	105	58	(10)	
	CDD18.3	1204	2	3	12	44	128	230	283	264	172	55	8	3	(11)	
	CDH23.3	11296	7	26	122	444	1180	2223	2911	2528	1371	411	56	18	(12)	
	CDH26.7	4727	0	2	19	117	441	1011	1379	1139	518	94	4	1	(13)	
(14) Wind	WSAvg	3.1	3.3	3.5	3.7	3.6	3.2	2.9	2.8	2.6	2.9	2.8	2.8	3.0	(14)	
(15) Precipitation	PrecAvg	1097	91	80	100	69	81	107	143	127	88	65	61	74	(15)	
	PrecMax	1498	178	174	278	156	208	236	283	365	223	275	208	166	(16)	
	PrecMin	747	18	21	9	6	12	15	38	37	0	0	10	11	(17)	
	PrecStd	185	45	41	55	40	43	53	62	74	55	61	43	43	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.3	26.6	28.8	31.6	34.9	36.7	37.2	37.3	34.2	31.1	27.2	25.7	(19)	
		MCWB	17.8	18.3	18.3	20.3	22.9	24.8	25.2	25.0	23.3	22.7	19.9	18.5	(20)	
	2%	DB	21.8	23.6	26.7	29.6	32.5	35.0	35.5	34.9	32.7	29.0	25.0	22.9	(21)	
		MCWB	16.7	16.7	17.5	19.6	22.1	24.3	25.1	24.9	23.4	21.4	18.3	17.9	(22)	
	5%	DB	19.5	21.2	24.4	27.9	31.0	33.4	34.1	33.4	31.4	27.5	23.0	20.6	(23)	
		MCWB	15.4	15.0	16.2	18.6	21.6	23.9	24.9	24.6	23.0	20.1	17.3	17.2	(24)	
	10%	DB	17.2	18.7	22.2	26.1	29.2	31.9	32.8	32.2	29.9	25.7	21.0	18.4	(25)	
		MCWB	13.8	13.4	15.1	17.8	20.8	23.4	24.7	24.3	22.6	19.8	16.7	15.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	19.9	20.3	22.7	24.6	26.5	27.2	26.9	25.7	25.1	21.7	20.9	(27)	
		MCDB	21.8	24.0	25.0	27.5	30.6	33.9	33.6	32.7	30.4	27.2	23.8	23.1	(28)	
	2%	WB	18.0	18.3	19.0	21.3	23.6	25.6	26.4	26.2	25.0	23.7	20.3	19.4	(29)	
		MCDB	20.7	22.2	23.7	26.6	29.4	32.1	32.8	31.8	29.7	26.8	23.1	21.5	(30)	
	5%	WB	16.3	16.6	18.0	20.4	22.8	25.0	25.8	25.6	24.4	22.4	19.0	17.7	(31)	
		MCDB	18.8	19.8	22.2	25.3	28.3	30.9	31.7	31.1	28.6	25.4	21.8	20.0	(32)	
	10%	WB	14.4	14.8	16.7	19.3	22.0	24.4	25.4	25.2	23.8	21.1	17.5	15.8	(33)	
		MCDB	16.7	17.9	20.7	24.0	27.1	29.7	31.0	30.3	27.6	24.3	20.3	18.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.9	12.7	13.0	12.0	10.7	10.3	10.1	10.2	11.7	12.5	11.2	(35)	
		MCDBR	13.4	14.5	14.9	14.6	13.4	12.3	11.9	11.7	12.0	12.9	14.0	12.9	(36)	
	5% WB	MCWBR	8.9	8.9	7.3	6.2	4.8	3.9	3.8	3.6	4.4	5.9	8.1	8.6	(37)	
		MCWBR	11.5	13.3	12.5	12.0	11.2	10.8	10.6	10.2	9.6	9.5	11.4	11.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.327	0.361	0.405	0.467	0.493	0.542	0.513	0.430	0.392	0.341	0.316	(40)	
		taud	2.501	2.472	2.406	2.270	2.152	2.123	2.006	2.078	2.299	2.374	2.475	2.545	(41)	
	Ebn at Noon		893	922	912	882	828	804	761	777	831	836	857	876	(42)	
		Edh at Noon	84	96	112	134	153	157	175	161	123	106	86	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.75	3.39	4.49	5.66	6.15	6.40	6.10	5.52	4.73	3.93	3.02	2.41	(44)	
	RadStd		0.15	0.35	0.36	0.39	0.48	0.44	0.38	0.27	0.40	0.47	0.24	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.38	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+188	+148	(46)	
(47) Regional (62 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+129	(47)	

Nomenclature: See separate page